uc3m | Universidad Carlos III de Madrid

Master Degree in Machine Learning for Health
2025-2026

Master Thesis

“Deep Learning for Photomultiplier Crystal Detection Signals”

Miguel Escudero Rodríguez

David Izquierdo-García

Leganés, September 2026

Deep Learning for Photomultiplier Crystal Detection Signals

Miguel Escudero Rodríguez
Master's Degree in Machine Learning for Health
Universidad Carlos III de Madrid, Leganés, Spain
Email: 100569281@alumnos.uc3m.es

***Abstract*—TODO: 150–250 palabras. Problema, metodo propuesto, datos empleados y resultados cuantitativos principales. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.**

Curabitur pretium tincidunt lacus. Nulla gravida orci a odio, sed fermentum lorem interdum. Phasellus convallis, sapien nec facilisis tincidunt, justo lectus malesuada erat, vitae consectetur neque arcu vel lorem. Integer consequat, augue a interdum consequat, erat sapien volutpat justo, at tincidunt libero mauris nec nisl. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Donec vitae augue sed justo interdum tincidunt.

Mauris euismod, lectus vitae tincidunt posuere, justo sem malesuada neque, eget feugiat sapien nisl a erat. Aenean interdum, purus vel malesuada consequat, magna sapien tincidunt augue, non consequat libero neque at justo. Suspendisse potenti. Proin vitae lorem at ipsum interdum tincidunt. Aliquam erat volutpat. Nam sed augue vel risus commodo interdum, sed tincidunt justo consequat. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Integer vel lorem nec sapien facilisis aliquam. Donec vulputate, magna at consequat suscipit, justo neque malesuada nisl, vitae tincidunt erat turpis vel ipsum. Vivamus feugiat lorem sed arcu malesuada, quis consequat purus tincidunt. Sed euismod, neque vitae interdum consequat, justo sapien fermentum mauris, vel tincidunt nulla nisl non erat.

***Index Terms*—Positron emission tomography, silicon photomultiplier, missing data imputation, deep learning, graph neural networks,silicon photomultiplier failure.**

1 Introduction

Positron Emission Tomography (PET) is a non-invasive molecular imaging modality and one of the central techniques of nuclear medicine, providing an *in vivo* assessment of the metabolic and biochemical state of a patient [1]. Unlike purely anatomical modalities, PET relies on radiotracers: biologically active molecules chemically labelled with positron-emitting radionuclides [2]. Once administered, the tracer distributes according to the physiological pathway targeted by its carrier molecule, so that the measured signal reflects function rather than structure. When an emitted positron annihilates with an electron in the surrounding tissue, two photons of 511 keV — the rest mass energy of the electron — are released in nearly opposite directions. Detecting both of them in coincidence defines a line of response (LOR) along which the annihilation took place, and the accumulation of millions of LORs allows the three-dimensional distribution of activity to be reconstructed [1]. This detection scheme gives PET a functional sensitivity that few other non-invasive imaging techniques can match [3], at the price of the tracer itself: positron-emitting radionuclides are radioactive by nature, costly to produce, and short-lived, which ties any PET facility to a nearby cyclotron and specific facilities [2].

This functional readout is most valuable where disease alters metabolism before it alters anatomy. In oncology, ^{18}F-FDG PET has become a cornerstone for diagnosis, staging, restaging and the assessment of treatment response [4], [5]. Its role in neurology is arguably even more distinctive, since it reveals pathologies that produce no macroscopic structural abnormality on conventional Magnetic Resonance Imaging (MRI) or Computed Tomography (CT), such as the characteristic hypometabolic and amyloid patterns of Alzheimer's disease [6], among other neurodegenerative and cerebrovascular disorders.

The sensitivity of PET is not matched by its spatial resolution. Clinical scanners typically resolve 4–5 mm full width at half maximum (FWHM) [7], and recent advances push the number towards 3 mm [7], [8], higher than the numbers normally achieved with MRI or CT. This limit decomposes into four contributions, two physical and two belonging to the detector, which behave very differently from one another [9]. The physical ones are the finite distance the positron travels before annihilating, set by the emitting isotope [10], and the small angular deviation from antiparallel emission, caused by the residual momentum of the electron–positron pair. Neither can be engineered away, although the second translates into a positional error that scales with the diameter of the detector ring and is therefore smaller in compact systems. The detector terms are more tractable. In pixelated arrays, the size of the individual element fixes the finest scale at which an interaction can be localised; shrinking it improves resolution, but the inter-crystal reflectors then occupy a growing fraction of the detector volume, and the probability that a photon deposits energy across several elements rises. The decoding error — the uncertainty in determining where within the module the interaction occurred — amounts to roughly one third of the detector width [9]. Unlike positron range and non-collinearity, this term is not a physical floor but a consequence of how the light distribution is read and interpreted, and it is therefore the one most open to improvement, whether through detector geometry, readout

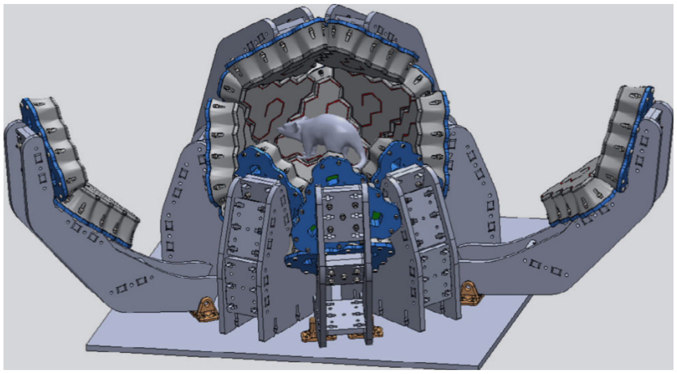

Figure 1: **TEMPORAL**Three-dimensional model of the quasi-spherical, hexagonal-based PET scanner. Reproduced from [12].

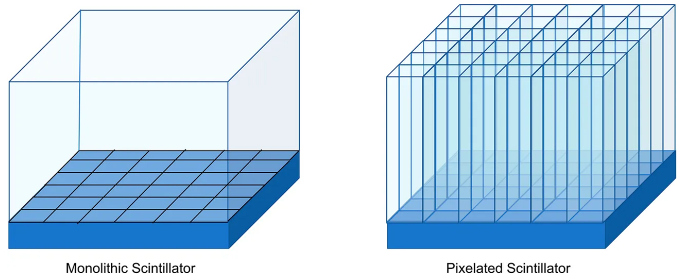

Figure 2: Monolithic (left) versus pixelated (right) scintillator configurations. In the monolithic design a continuous slab is coupled to a segmented photosensor array; in the pixelated design the crystal is cut into elements coupled one-to-one to the photosensors. Reproduced from [17] under CC BY 4.0.

electronics or the algorithms that interpret the signal. This is the ground occupied by organ-dedicated scanners. Adapting the detector geometry to a specific anatomical region increases solid-angle coverage, and with it sensitivity and resolution, at a lower manufacturing cost than a general-purpose whole-body system; designs exist for breast, brain, prostate and cardiac imaging [11]. The trade-off is that irregular geometries complicate calibration and data correction, and that small gantry apertures accentuate parallax errors and degrade the uniformity of spatial resolution across the field of view [11]. Much of the design effort in these systems therefore shifts onto the detector itself.

Our group is currently developing a quasi-spherical brain-dedicated PET scanner along these lines [12]. Rather than the cylindrical ring of conventional systems, its detector modules tile a closed surface around the head (Fig. 1), approaching full 4π steradian coverage and thereby collecting a far larger fraction of the emitted coincidences [12]. The modules are hexagonal, a shape chosen for two independent reasons: hexagons tile a surface without gaps and approximate a sphere more closely than square modules of comparable size, and hexagonal scintillators have also been shown to offer slightly better light yield and energy resolution than square, triangular or cylindrical prisms [13]. Each module is built around a monolithic LYSO crystal [14].

The choice between segmenting the scintillator and leaving it continuous is not incidental, and it conditions how the interaction position can be recovered at all [15]. In pixelated arrays, the crystal is cut into individual elements separated by reflectors, so that the interaction is assigned to whichever element fired. Event positioning is then simple and fast, but the resolution cannot be finer than the element itself [9], and two costs follow from making the elements smaller: the reflectors occupy a growing fraction of the detector volume, reducing the packing fraction, and the number of readout channels rises. Multiplexing schemes, in which several crystals share a readout line and the position is recovered by decoding the light distribution, are commonly used to keep this cost manageable [11]. Pixelated arrays also provide no intrinsic depth-of-interaction (DOI) information, which has motivated layered designs such as phoswich detectors, where stacked scintillators with different decay times allow a coarse depth assignment at the price of a degraded timing resolution [11].

Monolithic scintillators take the opposite approach. A single continuous block is coupled to an array of photodetectors, and the light from each interaction spreads over many channels at once [15]. Positioning is no longer a matter of identifying an element but of estimating a coordinate from the recorded light distribution, which shifts the resolution limit from the geometry of the crystal to the quality of the positioning algorithm, and makes the DOI recoverable as a continuous variable [16]. The absence of internal reflectors also restores the packing fraction lost in segmented designs [11]. This is the configuration adopted in the scanner described above, and it is the light sharing across the array that the present work exploits.

This light distribution is read out by an array of silicon photomultipliers (SiPMs) optically coupled to the rear face of the crystal [15]. Each interaction produces signal in a subset of channels, and a readout application-specific integrated circuit (ASIC) amplifies, thresholds and digitises each of them, so that an event is fully described by the vector of charges recorded across the array. Because a channel only produces an output once it exceeds the threshold set in the electronics, most entries of that vector are zero for any given event and only a few channels contribute; the sparsity of the signal is therefore a property of the acquisition system rather than a defect of the data [18].

Estimating the interaction position from that vector admits solutions of widely varying complexity. The most common is the centre of gravity (CoG), which places the event at the average of the photodetector coordinates, each weighted by the charge that channel recorded [19]; in practice the square of the charge is normally used as the weight, which reinforces the contribution of the channels closest to the interaction point and improves resolution [20]. Its appeal is that it is deterministic, has no parameters to tune and requires no prior calibration. In exchange, it exploits only part of the information contained in the light distribution: recovering the DOI, or correcting the

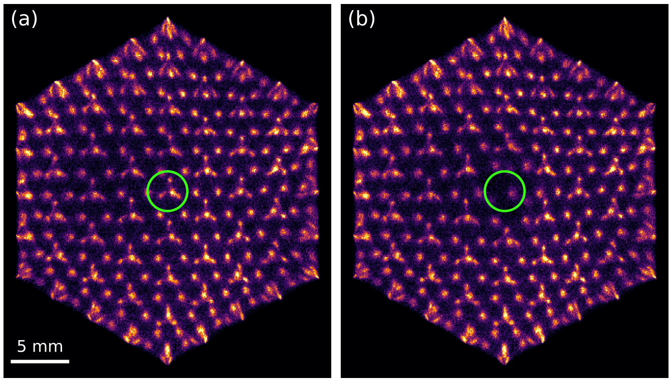

Figure 3: Flood map of the same detector module reconstructed by centre of gravity, using the same events in both panels: (a) with all 61 channels operational and (b) with one channel disabled, marked in green. The structures engraved in the crystal appear as discrete spots; the inset magnifies the neighbourhood of the disabled channel.

bias the method itself introduces near the crystal edges, calls for substantially more elaborate algorithms [21].

The two-dimensional histogram of the positions computed in this way over a large number of events forms the flood map (Fig. 3), in which the structures engraved in the crystal appear resolved as discrete spots. It is the image conventionally used to characterise a detector of this kind, and its sharpness depends on all 61 channels being operational: the CoG averages over the whole array, and a single missing term is enough to displace the centroid. The effect is confined to the neighbourhood of the failing channel — in the module of Fig. 3 the reconstructed position shifts by a median of 0.41 mm for events within 4 mm of it, and by less than 0.01 mm elsewhere — but within that neighbourhood the engraved structures are no longer resolved.

Silicon photomultipliers degrade and fail for a variety of reasons: thermal drift, manufacturing defects, deterioration of the optical coupling, or faults in the readout channel. The consequence for the image is well recognised in industrial practice: a dead pixel introduces reconstruction artefacts and incorrect activity values, and commercial systems include quality-control procedures to identify it [22]. The adopted general solution, however, is to discard the affected information and compensate for it downstream of positioning, either by duplicating events from neighbouring pixels in list mode or by interpolating in sinogram space [22]. The channel is not recovered: it is masked out.

This class of problem has motivated the application of deep learning to PET detector readout [23]. The most developed line is the estimation of the interaction position in monolithic crystals, where networks trained on calibration data consistently outperform analytical algorithms and recover the DOI as a continuous variable [24], [25]. Two previous studies address precisely this task on the present detector [18], [26]. All of them share the same framing: the light distribution recorded by the array is taken as input and the network returns a coordinate.

Closer to what is proposed here, deep learning has also been applied to recovering channel signals rather than positions. Networks have been trained to invert multiplexed readout schemes, reconstructing the individual channel values from a reduced set of measurements and validating the result on the recovered flood map [27], [28], and the approach has recently been extended to semi-monolithic detectors, where the full 64-channel response is reconstructed from 16 row-and-column sums by exploiting the light sharing between slabs [29]. In all these cases, however, the information carried by each channel is still present in the acquired data: multiplexing encodes it in a known linear combination, and the network learns to invert a mapping fixed by design. A failed channel is a different situation, since nothing of its signal reaches the acquisition and the only remaining trace of it is the statistical correlation that light sharing induces among its neighbours.

That situation is not unique to PET. Recovering the response of defective elements in a sensor array is an established problem in imaging, addressed either by interpolating from neighbouring pixels or, more recently, by training autoencoders to reconstruct corrupted regions from their surroundings [30]. The formulation adopted here belongs to that family: it is the masked autoencoder setting [31], in which part of the input is deliberately hidden and the model is trained to reconstruct it, which allows supervision to be generated from healthy modules without requiring labelled faulty detectors. What distinguishes the present case is that the redundancy being exploited is physical rather than statistical: the light emitted in a single interaction is measured simultaneously by several photodetectors, so the value of a channel is constrained by the response of those around it.

This work proposes to recover that channel rather than discard it. If the light from each interaction spreads over several photodetectors, the charge that the failed sensor would have recorded is not entirely lost: it remains implicit in what its neighbours measured, and the problem is to estimate it. The difficulty lies less in the model than in the supervision. In a faulty module there is no way of checking whether an estimate is correct, since the true charge of the dead channel is unobservable by definition: if it could be measured, the channel would not be faulty. What cannot be observed, however, can be simulated. It suffices to take a healthy module, disable one of its channels and keep the original value as a reference; the fault is fabricated, and with it the label. The whole approach rests on that idea, and it is applied to experimental data acquired with the detector described above rather than to simulations, so that the model learns the behaviour of the system as it actually is, with its noise, its threshold and its variation between modules.

The choice of model follows from the geometry of the array. The immediate option would be to treat the vector of sixty-one charges as a list of numbers and let a dense network discover on its own which channels are related to which, but that means learning from scratch something that is already known: which sensor is adjacent to which. The photodetectors form a hexagonal tiling in which every interior sensor has six immediate neighbours, and that neighbourhood is precisely the path along which the light is shared. A structure of this kind is naturally described as a graph, with one node per sensor and an edge between each pair of adjacent sensors, and it therefore admits a neural network that operates on it by propagating information between neighbours [32]. The model proposed here follows that approach, incorporating the geometry of the detector as prior knowledge instead of leaving it implicit in the data.

The evaluation is framed along the same lines. The error in the estimated charge is the quantity the model minimises, but on its own it says little about the usefulness of the method: what matters is how much is recovered of what the fault degrades. Three physical effects are therefore measured: the displacement of the reconstructed position, which is the direct consequence on the flood map; the conservation of the energy spectrum, which determines whether the restored signal is physically admissible; and the spatial resolution. All of this on modules the network has not seen during training, and compared against classical interpolation methods in order to establish that the complexity of the model is warranted. The behaviour of the method when more than one channel fails is also examined, with particular attention to contiguous groups, which are what is observed in real faulty detectors and also the least favourable case, since when an entire region disappears the lost sensor is left without healthy neighbours to draw on.

2 Materials and Methods

2.1 Detector and Data

Every measurement reported here comes from detector modules of the scanner described in Section 1. Each module is built around a monolithic LYSO hexagonal prism, 17.6 mm on a side and 13 mm thick, laser-etched into 364 optical structures that constrain how the scintillation light spreads before reaching the photosensor [14]. An array of silicon photomultipliers is optically coupled to its posterior face [12]. The array provides 61 instrumented readout channels, which tile a hexagonal lattice with a pitch of 3.75 mm and span 26 mm across, so that the sensors do not reach the crystal boundary. The lateral faces of the crystal carry an absorbing black adhesive that suppresses edge reflections. This has a consequence that recurs throughout our results: interactions near the perimeter lose a measurable fraction of their light, which shifts both the charge recorded by the peripheral channels and the position of the energy photopeak in those regions.

We used acquisitions recorded with a ^{22}Na source under flood irradiation, one file per physical module, comprising 159 healthy modules and 62 modules with at least one suspected faulty channel. Each file holds on the order of 10^6 events. An event activates 17 channels on average, ranging from a single channel to 54, since a channel only produces an output once it exceeds the threshold set in the readout electronics. We applied neither energy calibration nor an energy window. Both would require a per-channel calibration that is not available for this set of modules, and the metric we use to assess spectral fidelity is constructed so that their absence cancels, as Section 2.6 explains.

We split the healthy modules by file rather than by event, with a fixed seed: 149 for training, 5 for validation and 5 held out for testing. The distinction matters more than it might appear. Each file is a different physical detector, with its own gain, its own optical coupling and its own defects, so splitting by file measures generalisation to hardware the model has never seen. Splitting by event would let the model learn the idiosyncrasies of a detector and then be tested on that same detector, which is a considerably easier problem and not the one that arises in practice. The test partition was reserved before any training and used neither for fitting nor for model selection.

2.2 Preprocessing and Sample Generation

The raw quantity is a vector $\mathbf{r} \in \mathbb{R}^{61}$ holding the digitised charge of each channel for one event. We applied two operations before it reaches the network. Occasional negative values, produced by baseline subtraction in the electronics, were clipped to zero, since a negative charge has no physical meaning and would distort the centroid. Events in which no channel exceeded the threshold carry no information and were discarded.

We then simulated a faulty channel by forcing a subset $\mathcal{D}$ of the entries to zero and asking the model to recover the original vector from what remained. The network additionally receives a binary mask indicating which entries were suppressed, so that a channel switched off is distinguishable from a channel that legitimately recorded no signal. This is the masked autoencoder setting [31] applied to detector channels instead of image patches, and it makes the supervision self-generated: the target is the original vector, so no module with a genuinely dead sensor is needed for training. We drew a fresh $\mathcal{D}$ for every sample, so a given event is never seen twice with the same failure pattern.

Normalisation, however, deserves a comment, because it constrains the architecture. We scaled each event by the maximum of its own masked vector,

$$\mathbf{x} = \frac{\tilde{\mathbf{r}}}{\max_c \tilde{r}_c}, \qquad \mathbf{y} = \frac{\mathbf{r}}{\max_c \tilde{r}_c}, \tag{1}$$

where $\tilde{\mathbf{r}}$ is the masked vector and $\mathbf{y}$ the target. Two choices are embedded here. Scaling per event rather than per dataset follows from the physics: for position estimation what matters is how the light distributes across the array, not how much light the interaction released. Computing the factor *after* masking means that the target of a suppressed channel exceeds unity whenever that channel was the brightest of the event. The output layer is therefore linear, with no sigmoid or clamp, because bounding it would make the brightest failures unrecoverable by construction.

One further choice concerns which channel to suppress. Roughly three quarters of the channels in any given event are below threshold, so drawing the seed uniformly would produce a training set in which most samples require no correction at

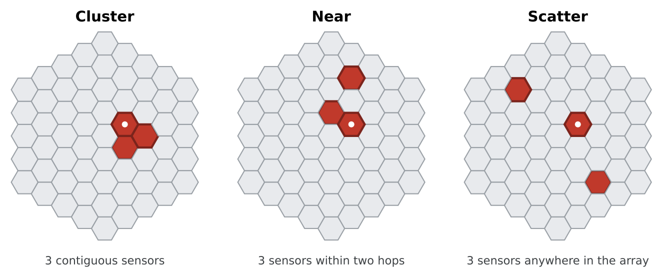

Figure 4: The three failure regimes used to generate the training masks, shown on the 61-sensor array for $k = 3$. Suppressed sensors are marked in red and the white dot indicates the seed from which the pattern grows. Cluster forces the failures to be contiguous, scatter distributes them across the array, and near places them within two hops of the seed without requiring contact.

all, and the model would reasonably learn to predict zero. We therefore balanced the draw: half of the samples suppress a channel that carried signal and half suppress a channel already at zero.

2.3 Failure Regimes

A single dead channel is the simplest fault, but not the only one. When we inspected the flood maps of the 62 defective modules, we found three recurring patterns. What matters is not so much how many sensors fail as how they are arranged: one whose neighbours are all healthy can be reconstructed from them, whereas one surrounded by other dead sensors cannot.

We reproduced those patterns with three mask-generation strategies, illustrated in Fig. 4. In the *cluster* regime the k suppressed channels grow outward from the seed so that they remain contiguous, so an entire region of the array disappears at once and the affected sensors are left without healthy immediate neighbours. In the *scatter* regime they are drawn uniformly across the array, so each failure is effectively isolated. The *near* regime sits between the two: the k channels lie within two hops of the seed without being required to touch. That intermediate case is the one we observed most often in real faulty modules, typically two or three sensors separated by one or two positions, and it is the reason we introduced it rather than settling for the two extremes.

2.4 Network Architecture

The geometry of the array suggests how to model the problem. A dense network receiving the 61 charges as a flat vector would have to learn which sensors are adjacent, something already known from the physical layout. We therefore represented the detector as a graph whose nodes are the 61 active sensors and whose edges join physically adjacent ones, and propagated information along those edges [32], [33]. The resulting adjacency has 312 directed edges, with 37 nodes of degree six, 18 of degree four and 6 of degree three, and it is fully symmetric.

Each layer of our model, which we refer to as HexGNN, updates a node from its own state and the mean of its neighbours,

$$h_c^{(l+1)} = \sigma\left(W_s h_c^{(l)} + W_n \frac{1}{|\mathcal{N}(c)|} \sum_{j \in \mathcal{N}(c)} h_j^{(l)} + b \right), \tag{2}$$

where $\mathcal{N}(c)$ are the neighbours of sensor c and σ is a Gaussian error linear unit (GELU) [34]. The aggregation is isotropic: every neighbour of a node shares the weight matrix W_n, so the layer encodes adjacency but not direction. Batch normalisation [35] is applied between blocks. Our reference configuration uses a hidden width of 48 and four blocks, for 38 305 trainable parameters.

We compared this model against three groups of alternatives. As classical baselines without learned parameters we used the mean of the neighbouring channels and a per-channel linear regression on the remaining 60, both fitted on training modules only and both excluding the target channel from their inputs. As neural baselines without a geometric prior we trained a deep multilayer perceptron and a residual multilayer perceptron with channel attention. Both receive the same information but no notion of adjacency, which isolates the contribution of the prior from that of raw capacity.

The aggregation in Eq. 2 is deliberately simple, and it is fair to ask whether that simplicity is what limits performance. We therefore tested four alternatives, each addressing a different suspicion about where the bottleneck might be.

If the six neighbours of a sensor contribute unequally — and they might, since the light reaching one of them depends on the direction of the interaction — then averaging them discards useful information. Graph attention networks (GAT) [36] learn a weight per neighbour instead of fixing it, which lets the model decide how much each one should count. A related but broader concern is that the mean is only one summary of a neighbourhood. Principal neighbourhood aggregation (PNA) combines several statistics and scales them by node degree, retaining information that any single one throws away [37]; degree varies here between three and six, so the scaling is not vacuous.

The other two alternatives question the locality of the model rather than its aggregation. Our layers propagate information one hop at a time, so with four blocks a sensor only sees four rings away. If the missing charge were constrained by the global shape of the light distribution, a wider receptive field would help. We tested this in two forms: a graph U-Net, which pools the graph into a coarser representation before expanding it back [38], and a graph transformer (Graphormer) in which every node attends to every other one regardless of adjacency [39]. We also tested metric edge features, anisotropic hexagonal convolution with one weight per direction, per-channel equalisation, heteroscedastic prediction of mean and variance, and averaging ensembles [40].

Table 1: Training configuration of the reference model.

Parameter	Value
Architecture	HexGNN, 4 blocks, hidden width 48
Trainable parameters	38 305
Activation	GELU
Loss	MSE over the 61 channels
Optimiser	AdamW [42]
Learning rate	10^{-3}, cosine decay [43]
Weight decay	10^{-4}
Batch size	512
Training samples	15 943 000 (single pass)
Validation events	400 000, fixed masks
Model selection	Validation MAE on the imputed channel

2.5 *Training*

We trained with the mean squared error (MSE) computed over the full 61-channel output rather than the suppressed entries alone,

$$\mathcal{L} = \frac{1}{61}\sum_{c=1}^{61}\left(\hat{y}_c - y_c\right)^2 , \tag{3}$$

which treats the task as ordinary regression and keeps the healthy channels as a consistency constraint. We selected this loss after comparing it against the mean absolute error (MAE) and the Huber loss [41] with $\delta = 0.1$. We also tested two physics-informed variants, one penalising the displacement of the reconstructed position and one penalising the shift of the energy spectrum; both are reported in Section 3.

Table 1 lists the configuration. Two aspects of the procedure are worth making explicit. First, the training set does not fit in memory, so modules are loaded one at a time and contribute 107 000 events each; the model therefore performs a single pass over 15 943 000 masked samples. Since the masks are regenerated for every sample, no exact input is ever repeated, which is why we applied no early stopping. Second, we selected the final weights by the validation error on the imputed channel, computed on a fixed set of 400 000 events with a fixed mask seed. We did not use the total loss for this purpose. It is dominated by the 60 healthy channels and stays nearly flat across the run, so it separates checkpoints poorly.

Each run stores its preprocessing configuration inside the checkpoint. This is a safeguard rather than a detail: a model trained with one normalisation and evaluated with another produces meaningless numbers without raising any error, and verifying the match automatically removes that failure mode.

2.6 *Evaluation Metrics*

The quantity the model minimises is the error in the recovered charge, but on its own it says little about whether the method is useful. What matters is how much of the degradation caused by the fault is undone. We therefore evaluated on three levels: the charge itself, the position it produces, and the physical quantities the detector is built to deliver.

At the charge level we report, over the N events in which channel c was suppressed, the MAE and the bias

$$\mathrm{MAE}_c = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|, \qquad \mathrm{bias}_c = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i), \tag{4}$$

both in the normalised units of Eq. 1. The bias acts as a guardrail. A model can attain a low MAE while predicting systematically low, which would displace every reconstructed position in the same direction; only the signed average reveals it. Because a normalised error is hard to interpret on its own, we also express the MAE relative to the mean charge that the channel collects. That ratio is sensitive to the number of events used for the denominator, so we computed numerator and denominator on the same sample throughout.

The position level is our primary metric, since it is the one that reaches the image. For each event we computed the interaction position by CoG weighted by the squared charge [19], [20], in three versions of the same event: intact, with the channel suppressed, and with the channel imputed. Comparing them against the intact position gives two displacements. The *degraded* displacement ΔR_{deg} is how far the centroid moves when the fault goes uncorrected, and therefore measures the damage. The *imputed* displacement ΔR_{imp} is how far it still lies from the true position after the model has acted, and therefore measures the residual error. Recovery is the fraction of the damage the model removes,

$$\mathrm{Recovery} = 100 \times \left(1 - \frac{\Delta R_{\mathrm{imp}}}{\Delta R_{\mathrm{deg}}}\right), \tag{5}$$

with both terms evaluated at the same quantile and over the same population of events. A value of 100 % would mean the imputed position coincides with the intact one, and 0 % that imputing changed nothing. We report the value at the 90th percentile, which weights the difficult events, and also the mean. Averaging over the 61 channels gives the macro figures used throughout, and the minimum over channels characterises the worst case. Since a ratio says nothing about the magnitude of what remains, we also report ΔR_{imp} in millimetres.

Two further analyses ask a different question. Recovery tells us how close the imputed position is to the true one, but not whether the restored signal is physically admissible. A model could place events correctly on average while distorting the quantities the detector is built to measure.

The first analysis characterises the *spread* of the positioning error rather than its typical size. Recovery reports a percentile of ΔR; here we summarise the same underlying distribution by its width, taking the error components Δx and Δy and computing a Gaussian-equivalent full width at half maximum from a robust scale estimate,

$$\mathrm{FWHM} = 2.3548 \cdot \tfrac{1}{2}\left(\hat{\sigma}_{\Delta x} + \hat{\sigma}_{\Delta y}\right), \qquad \hat{\sigma}_v = 1.4826 \cdot \mathrm{MAD}(v), \tag{6}$$

where MAD denotes the median absolute deviation, used so that a few extreme events do not drive the estimate. Alongside it we report the bias $\|(\overline{\Delta x}, \overline{\Delta y})\|$, which separates a systematic displacement of the map from a symmetric blurring around the correct position.

We stress that this is not the spatial resolution of the detector. It does not come from fitting the spots of the flood map, and it is measured on the error that the fault introduces, not on the point-spread function of the system; values of the two quantities differ by about two orders of magnitude and are not comparable. Nor is it independent of the recovery of Eq. 5: both summarise the same distribution of positioning errors, one by a percentile and the other by its width, and we report both because a method could reduce the typical error while leaving a long tail, or the reverse.

The second concerns energy. Suppressing a channel removes part of the collected light, so the reconstructed energy of the event falls and the photopeak shifts towards lower values; a good imputation should put it back. Measuring this is complicated by the absence of energy calibration, since we cannot convert ADC counts into keV. We avoided the problem with a paired differential design: over the same events we subtract the true charge of the channel and add back the prediction, so any calibration factor multiplies both terms and cancels. We evaluated this by zones of the crystal, defined as concentric hexagonal bands, because the absorbing adhesive on the lateral faces makes the periphery behave differently from the centre [21] and an average over the whole crystal would hide that difference.

Finally, a note on what these numbers can and cannot support. Comparisons between models are paired, in that every model is evaluated on the same modules, the same events and the same masks, so relative differences are measured precisely. Absolute performance on a new detector, however, is a different matter, because modules differ from one another far more than training runs do. We quantified both sources of variation separately, and Section 3 reports them alongside the results they qualify.

3 Results

Unless stated otherwise, every figure in this section refers to the reference model evaluated on the five held-out modules. Results are quoted as mean $\pm$ standard deviation over independent training runs, and we state the sample size in each case. Evaluation campaigns differ slightly in the number of events drawn per test file. We measured that effect by evaluating the same model under two of them and found it below 0.3 percentage points, well under the variability reported here, so the text does not distinguish between them.

3.1 Imputation Accuracy

The model recovers the charge of a suppressed channel with a mean absolute error of 0.6228 ± 0.0064 in normalised units ($n = 4$). Expressed relative to the mean charge that each channel collects, this amounts to a 29.1 % error, and it is not uniform across the array: 25.9 % for channels in the core of the crystal against 34.0 % at the edge. The absorbing adhesive on the lateral faces accounts for that gap, since peripheral channels collect less light and their neighbours carry correspondingly less information about them.

Table 2: Displacement of the reconstructed position with respect to the intact event, before and after imputation, in millimetres.

	Degraded	**Imputed**	**Recovery**
Median	0.099	0.047	52.1 %
Mean	0.241	0.118	51.0 %
90th percentile	0.666	0.288	56.8 %

The bias stays at $-0.062{\pm}0.023$, that is, below a tenth of the MAE. This matters because the failure mode we were guarding against is a model that minimises its error by predicting conservatively low values, which would look acceptable in the MAE and would systematically displace the reconstructed position. It does not happen.

3.2 Recovery of the Reconstructed Position

Charge error is an intermediate quantity. What reaches the image is the displacement of the centroid, and Table 2 gives it in millimetres. Suppressing a channel moves the reconstructed position of the affected events by a median of 0.099 mm and, for the worst tenth of them, by 0.666 mm. After imputation those figures fall to 0.047 mm and 0.288 mm. The residual displacement at the 90th percentile is therefore 7.7 % of the 3.75 mm sensor pitch.

Fig. 5 shows what this means for the image. Each panel is the difference between a flood map and the intact reference, averaged over failures of the sensors along one line of the array. Without correction the difference covers the neighbourhood of the failing channel with a pattern of excess and deficit counts, which is the signature of events being displaced towards the surviving sensors. After imputation that pattern largely disappears. The lower row, which follows a line towards the edge of the crystal, shows a residual that the upper row does not: the periphery is where the method has least information to work with, in agreement with the per-channel errors of Section 3.1.

Performance is not uniform across the detector. Fig. 6 maps the recovery of each of the 61 channels and reproduces the same core-to-edge gradient. Central channels, whose light is shared with six neighbours, exceed 60 %; channels on the perimeter, with three or four neighbours and less collected light, fall below 45 %.

Fig. 7 closes the sequence with the flood map itself, where the practical consequence is visible: the etched structures around the failing channel, which the fault blurs into one another, are resolved again after imputation.

3.3 Value of the Geometric Prior

Table 3 compares the architectures under a common campaign. The two models without a geometric prior reach 51.39 % and 48.92 % recovery at the 90th percentile. The smallest graph model reaches 58.0 % with an order of magnitude fewer parameters than either of them.

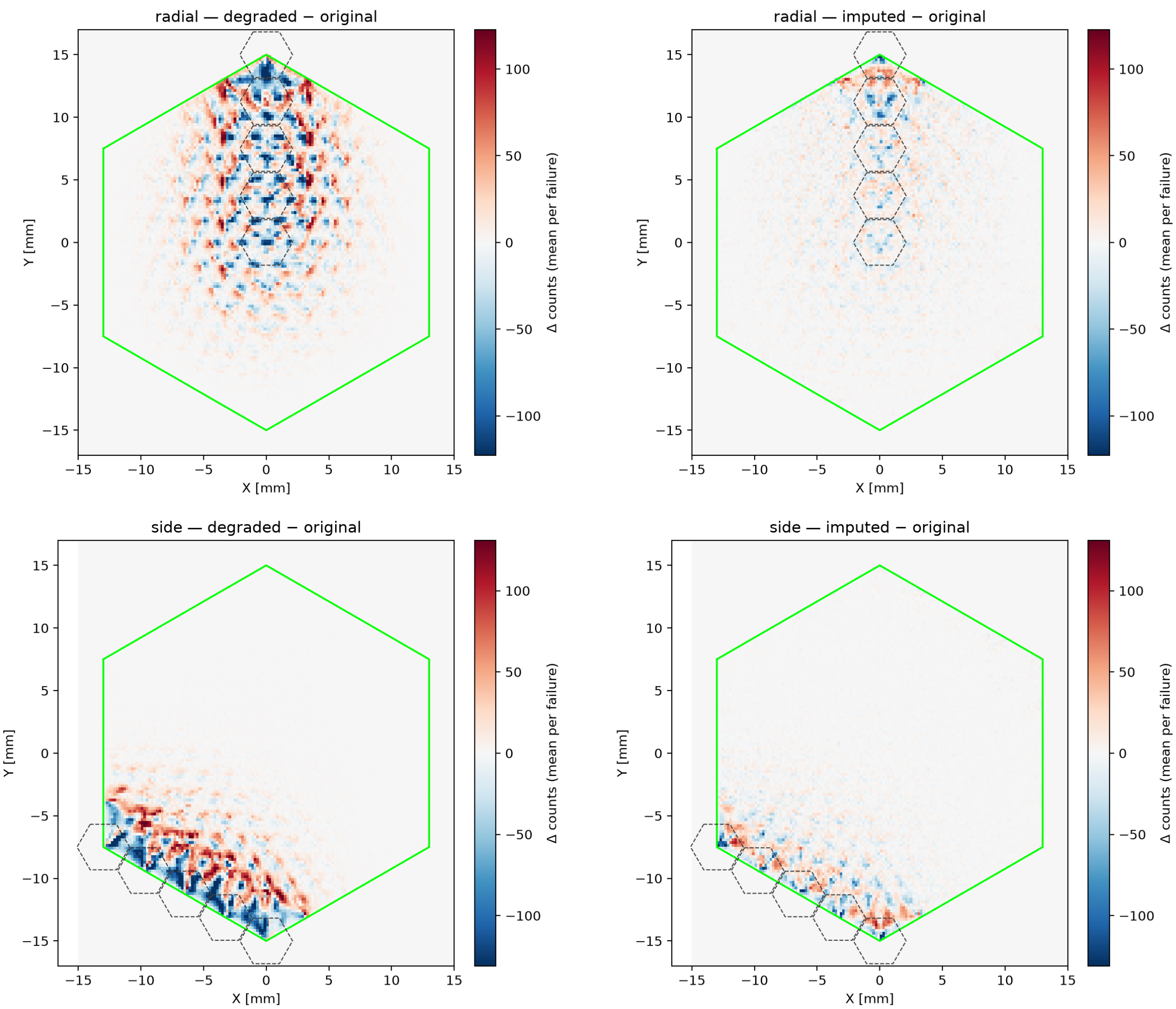

Figure 5: Average flood-map distortion caused by the failure of a sensor. Each panel shows the difference in counts with respect to the intact map: red where the fault adds events and blue where it removes them. Top row, sensors along a radial line; bottom row, sensors along a line towards the edge. Left, without correction; right, after imputation. The dashed hexagons mark the sensors involved and the green outline the crystal boundary.

Table 3: Architecture comparison under a common evaluation campaign. Recovery at the 90th percentile and on average, in per cent; MAE in normalised units.

Model	**Params**	**Rec. p90**	**Rec. mean**	**MAE**
DeepMLP	500 541	51.39	45.90	0.7219
ResidualMLP	345 981	48.92	44.81	0.7299
Graphormer	4 263 881	55.17	49.65	0.6600
Graph U-Net	331 649	55.22	47.95	0.6918
PNA	165 761	58.45	52.48	0.6309
HexGNN-s	38 305	58.02	52.66	0.6195
HexGNN-m	225 217	58.27	52.64	0.6185
HexGNN-l	398 593	57.47	51.96	0.6385

Neither capacity nor a more elaborate aggregation moves that figure. Widening the graph model changes recovery by less than a point in either direction. The two architectures with a wider receptive field, however, perform three points *worse* despite using up to a hundred times more parameters, and combining several aggregation statistics matches the reference rather than exceeding it. The problem behaves as a local one, and the isotropic mean over immediate neighbours is enough to solve it.

To test whether the advantage comes from the detector geometry rather than from message passing in general, we retrained the reference with the adjacency randomly permuted, preserving the number of nodes, the number of edges and the degree distribution. The graph is then equally connected but physically meaningless. Recovery collapses to 28.08 %,

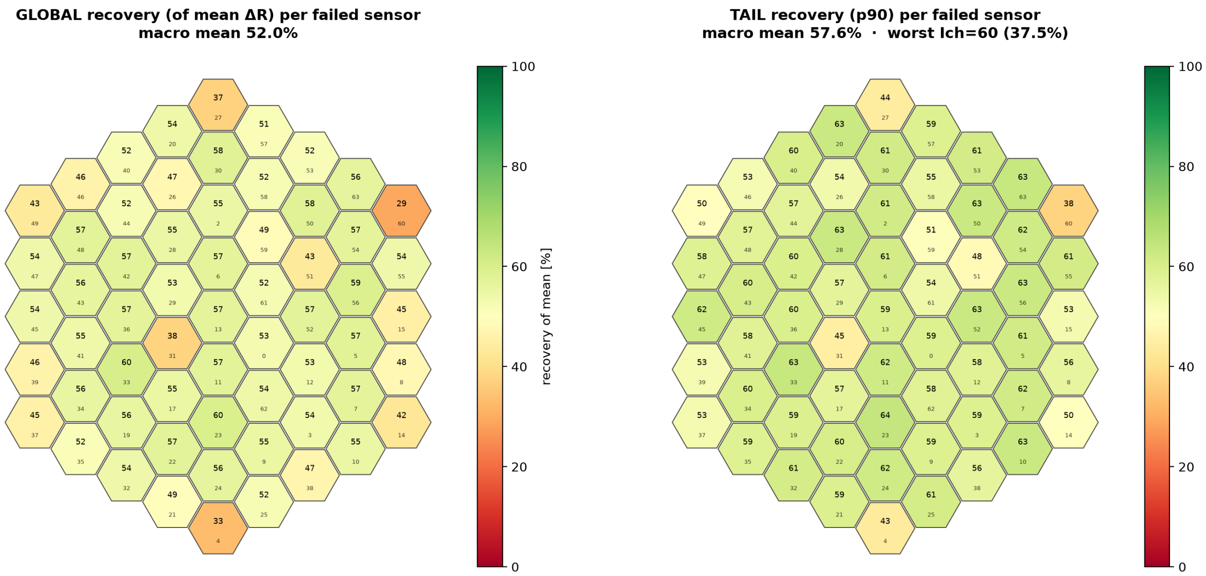

Figure 6: Recovery per channel over the 61 active sensors, at the 90th percentile (left) and on average (right). The colour scale is common to both panels. Recovery degrades towards the perimeter of the array.

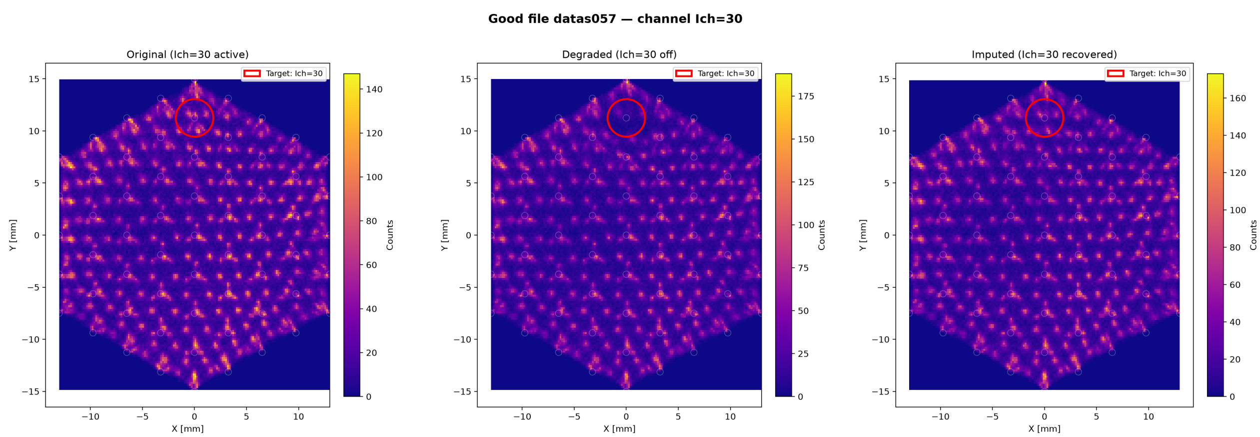

Figure 7: Flood map of a test module reconstructed from the same events. Left, with all 61 channels operational; centre, with one channel disabled; right, after imputing it.

27.66 % and 31.28 % across three independent permutations, roughly half of the 58.0 % obtained with the true adjacency. The prior therefore accounts for about thirty points of the sixty the model achieves, and it is the specific adjacency of the detector that matters, not the fact of using a graph.

Against methods without learned parameters the network recovers 57.9 %, compared with 45.41 % for the mean of the neighbouring channels and 46.56 % for a per-channel linear regression on the remaining 60. The margin is 12.49 and 11.33 percentage points respectively, which answers the question of whether the complexity of the model is warranted.

The linear baselines also let us measure how far the useful information extends. Restricting the regression to the first ring of neighbours, 5.1 channels on average, gives 44.04 % recovery. Adding the second ring, 13.7 channels, raises it to 46.42 %. The third ring adds 0.09 points and using all 60 remaining channels adds a further 0.04. Beyond the second ring there is essentially nothing left to extract, which is consistent with a model whose layers propagate information only a few hops.

3.4 Generalisation and Variability

A single figure of merit says little without knowing how much it moves, so we measured the two sources of variability separately (Table 4). Across four independent training runs on a fixed partition, differing in the global seed and in the order of the modules, recovery at the 90th percentile is 58.04 ± 0.37 %.

Repeating the experiment with five different partitions of the same 159 modules, however, gives 56.56 ± 2.32 %, a standard deviation six times larger. Evaluating the model on individual unseen modules explains why: recovery ranges from 29.2 % to

Table 4: Sources of variability in the reference model. Values are mean $\pm$ standard deviation over n independent runs.

Source of variation	n	**Rec. p90**	**Rec. mean**
Training run, fixed partition	4	58.04 ± 0.37	52.15 ± 0.31
Data partition	5	56.56 ± 2.32	50.60 ± 2.04

Table 5: Recovery at the 90th percentile by failure regime, averaged over $k = 1 \ldots 4$ suppressed channels, over a campaign of 20 mask seeds.

Training regime	**Cluster**	**Near**	**Scatter**
Single channel (n=3)	33.74 ± 11.68	38.28 ± 9.23	47.69 ± 4.95
Cluster, k=1−4 (n=1)	58.46	52.62	52.44
Near, k=1−4 (n=3)	58.31 ± 0.51	54.61 ± 0.58	53.43 ± 0.67

63.0 % depending on which detector it is measured on, with a standard deviation of 7.86 percentage points across modules. With five modules in the test partition, that places the standard error of the reported mean at 3.5 points.

The consequence is worth stating plainly, because it conditions how every other number here should be read. Comparisons between models are paired and therefore precise: the same modules, the same events and the same masks, with differences of tenths of a point resolvable. Absolute performance on a detector that has never been measured is far less certain, and the dominant term is not the training procedure but which detectors happen to fall in the test partition. We therefore take 56.56 ± 2.32 % as the figure that describes what the method achieves on unseen hardware.

The same caution applies to the worst channel of the array, which we had initially considered as a robustness figure. Over four independent runs it averages 42.29 ± 4.45 %, spanning 37.54 % to 47.76 %, and the channel attaining that minimum is not even the same in every run. As a single-point statistic over 61 channels it is too noisy to support any claim, and we do not use it as such.

3.5 Multiple Simultaneous Failures

Training with a single suppressed channel does not transfer to multiple failures. Table 5 reports recovery averaged over $k = 1 \ldots 4$ suppressed channels in the three regimes of Section 2.3. A model trained on single failures drops to 33.74 % on contiguous ones, and its standard deviation across runs rises to 11.68 points because it performs adequately at $k = 1$ and collapses at $k = 4$. Models trained with k drawn from 1 to 4 stay between 52 % and 59 % in every regime, with standard deviations below 0.7 points.

The gap between the first row and the other two is the largest effect we measured: 24.6 points on contiguous failures and 16.3 on the intermediate regime. Exposing the model to multiple failures during training is what determines whether it survives them, and the cost of doing so is negligible, since the multi-failure models lose nothing at $k = 1$.

Which regime is used for that training matters far less. The model trained on the intermediate regime exceeds the

Table 6: Recovery and residual displacement by regime as the number of suppressed channels grows, for the model trained on the intermediate regime ($n = 3$). Displacements at the 90th percentile, in millimetres.

	Recovery (%)			**Residual ΔR_{imp} (mm)**		
k	Clus.	Near	Scat.	Clus.	Near	Scat.
1	56.2	56.2	56.2	0.310	0.310	0.310
2	**59.5**	55.4	55.2	0.464	0.375	**0.305**
3	**59.9**	53.5	51.6	0.532	0.444	**0.366**
4	**57.7**	53.3	50.7	0.628	0.529	**0.366**

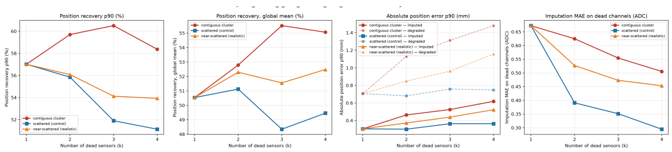

Figure 8: Recovery as a function of the number of simultaneously suppressed channels, for a model trained on single failures and one trained on one to four.

one trained on contiguous clusters by 1.99 points in its own regime, a difference of 2.9 standard deviations, and the two are indistinguishable on contiguous failures (−0.14 points) and on scattered ones (+0.99 points, 1.3 standard deviations).

The behaviour across regimes deserves care, because the relative and absolute readings point in opposite directions (Table 6). In recovery, contiguous failures are the *easiest* case for every $k \geq 2$: 57.7 % at $k = 4$ against 50.7 % for scattered ones. In millimetres they are the worst: a contiguous group of four displaces the centroid by 1.480 mm at the 90th percentile, twice as much as four scattered channels, and after imputation still leaves 0.628 mm against 0.366 mm. Both statements come from the same runs. Recovery is a ratio, and a fault that causes more damage offers more damage to undo, so a larger fraction repaired can coexist with a larger residual error. Scattered failures are the least damaging in absolute terms and, at the same time, the hardest to correct proportionally.

3.6 Physical Fidelity

Recovering the position is not the same as restoring a physically admissible signal, so we measured the two quantities the detector is built to deliver.

The spread of the positioning error, measured as the Gaussian-equivalent FWHM of Eq. 6, narrows from 0.1656 mm with a suppressed channel to 0.0872 ± 0.0015 mm after imputation ($n = 3$, Fig. 9), so imputation removes 47.3 ± 0.9 % of the blur the fault introduces. The bias behaves better than the width: it falls from 0.1262 mm to 0.0220 ± 0.0020 mm, a reduction of 82.6 ± 1.6 %. The error distribution after imputation is therefore almost centred on the true position, and what remains is a symmetric spread rather than a systematic displacement.

The energy spectrum is preserved to a comparable degree, with a global spectral recovery of 84.7 ± 0.9 % ($n = 3$, Fig. 10). Here the core-to-edge gradient reappears: $87.4 \pm$

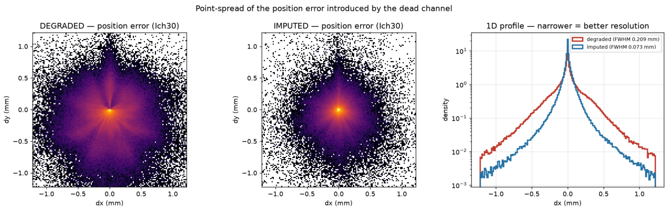

Figure 9: Distribution of the positioning error with one channel suppressed and after imputation. The width of each distribution is the quantity reported as FWHM in Eq. 6.

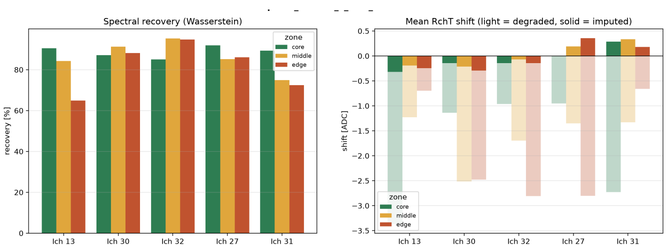

Figure 10: Energy spectrum by crystal zone for the original, degraded and imputed events. Zones are concentric hexagonal bands from the centre to the perimeter.

$1.8\,\%$ in the core, $85.6 \pm 0.7\,\%$ in the middle band and $81.2 \pm 0.3\,\%$ at the edge. The residual shift of the photopeak stays at 0.21 ± 0.005 ADC units, an order of magnitude below the shift the fault introduces.

3.7 Variants That Did Not Improve the Model

Several of the alternatives described in Sections 2.4 and 2.5 failed to improve on the reference, and the reasons are informative.

Adding the position error to the loss changes recovery from $58.0\,\%$ to $58.2\,\%$, within the run-to-run band. This is not surprising in hindsight: charge and position are related monotonically for a given channel, so the term is largely redundant with the MSE already being minimised.

Adding a term on the energy spectrum is actively harmful, dropping recovery to $46.0\,\%$ and inverting the sign of the bias. The cause is the form of the term rather than the quantity it targets. Spectral agreement is computed over a batch, so the penalty acts on a signed average, and the model can reduce it by letting errors of opposite sign cancel between events. It optimises the aggregate while degrading every individual event, which is precisely what the per-event metrics detect. Any physics-informed term for this problem needs to be computed per event.

Predicting a variance alongside the mean, trained with a Gaussian negative log-likelihood, reaches $20.1\,\%$. The likelihood lets the model reduce its loss by declaring large uncertainty instead of improving the prediction, and with no constraint tying the variance to a calibration target it takes that route.

Ensembling, however, does help, if modestly. Averaging three independently trained replicas raises recovery to $58.69\,\%$ and averaging three different architectures to $58.73\,\%$, against $58.04 \pm 0.37\,\%$ for a single model. Both are the only variants we found that exceed the reference band, and the gain is around 0.7 points at three times the inference cost.

4 Discussion

Interpretacion, comparacion con el estado del arte y limitaciones.

5 Conclusions

Conclusiones y lineas futuras.

References

[1] S. R. Cherry, J. A. Sorenson, and M. E. Phelps, *Physics in Nuclear Medicine*, 4th ed. Philadelphia, PA: Elsevier Saunders, 2012.

[2] J. Rong, A. Haider, T. E. Jeppesen, L. Josephson, and S. H. Liang, “Radiochemistry for positron emission tomography,” *Nature Communications*, vol. 14, 2023.

[3] J. J. Vaquero and P. Kinahan, “Positron emission tomography: Current challenges and opportunities for technological advances in clinical and preclinical imaging systems,” *Annual Review of Biomedical Engineering*, vol. 17, pp. 385–414, 2015.

[4] J. W. Fletcher, B. Djulbegovic, H. P. Soares *et al.*, “Recommendations on the use of ^{18}f-fdg PET in oncology,” *Journal of Nuclear Medicine*, vol. 49, no. 3, pp. 480–508, 2008.

[5] R. Boellaard, R. Delgado-Bolton, W. J. G. Oyen *et al.*, “FDG PET/CT: EANM procedure guidelines for tumour imaging: Version 2.0,” *European Journal of Nuclear Medicine and Molecular Imaging*, vol. 42, no. 2, pp. 328–354, 2015.

[6] K. A. Johnson, S. Minoshima, N. I. Bohnen *et al.*, “Appropriate use criteria for amyloid PET,” *Journal of Nuclear Medicine*, vol. 54, no. 3, pp. 476–490, 2013.

[7] B. A. Spencer, E. Berg, J. P. Schmall *et al.*, “Performance evaluation of the uEXPLORER total-body PET/CT scanner based on NEMA NU 2-2018,” *Journal of Nuclear Medicine*, vol. 62, no. 6, pp. 861–870, 2021.

[8] A. González-Montoro, J. Barberá, D. Sánchez, Á. Mondéjar, M. Freire, K. Díaz, A. Lucero, S. Jiménez-Serrano, J. Alamo, C. Morera-Ballester, J. Barrio, N. Cucarella, V. Ilisie, L. Moliner, C. Valladares, A. J. González, J. Prior, and J. M. Benlloch, “A new brain dedicated PET scanner with 4D detector information,” *Bio-Algorithms and Med-Systems*, vol. 18, no. 1, pp. 107–119, 2022.

[9] W. W. Moses, “Fundamental limits of spatial resolution in PET,” *Nuclear Instruments and Methods in Physics Research A*, vol. 648, no. Supplement 1, pp. S236–S240, 2011.

[10] C. S. Levin and E. J. Hoffman, “Calculation of positron range and its effect on the fundamental limit of PET system spatial resolution,” *Physics in Medicine and Biology*, vol. 44, no. 3, pp. 781–799, 1999.

[11] A. Sanaat, M. Amini, H. Arabi, and H. Zaidi, “The quest for multifunctional and dedicated PET instrumentation with irregular geometries,” *Annals of Nuclear Medicine*, vol. 38, no. 1, pp. 31–70, 2024.

[12] D. Pérez-Benito, R. Chil, J. M. Udías, M. Desco, and J. J. Vaquero, “SiPM-based PET detector module for a 4π span scanner,” vol. 936, pp. 18–21.

[13] D. Perez-Benito, R. Chil, L. A. Hidalgo-Torres, and J. J. Vaquero, “Scintillator geometrical considerations for detectors based on hexagonal SiPMs,” vol. 7, no. 7, pp. 684–691. [Online]. Available: https://ieeexplore.ieee.org/document/10133867/

[14] D. Perez-Benito, L. Batres, M. Vidal, and J. J. Vaquero, “Laser-microstructured hexagonal detectors for high-sensitivity quasi-spherical brain PET,” in *2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD)*, pp. 1–1. [Online]. Available: https://ieeexplore.ieee.org/document/11287358/

[15] A. González-Montoro, M. N. Ullah, and C. S. Levin, “Advances in detector instrumentation for PET,” *Journal of Nuclear Medicine*, vol. 63, no. 8, pp. 1138–1144, 2022.

[16] S. España, R. Marcinkowski, V. Keereman, S. Vandenberghe, and R. Van Holen, “DigiPET: sub-millimeter spatial resolution small-animal PET imaging using thin monolithic scintillators,” vol. 59, pp. 3405–3420.

[17] E. Enlow and S. Abbaszadeh, "State-of-the-art challenges and emerging technologies in radiation detection for nuclear medicine imaging: A review," *Frontiers in Physics*, vol. 11, p. 1106546, 2023.

[18] L. Jiménez Algarra, "Deep learning for precise gamma-ray interaction in a hexagonal monolithic PET detector," Master's thesis, Universidad Carlos III de Madrid, Leganés, Spain, 2025.

[19] H. O. Anger, "Scintillation camera," vol. 29, no. 1, pp. 27–33.

[20] H. Poladyan, O. Bubon, A. Teymurazyan, S. Senchurov, and A. Reznik, "Gaussian position-weighted center of gravity algorithm for multiplexed readout," vol. 65, no. 16, p. 165003.

[21] P. Conde, A. Iborra, A. J. González, L. Hernández, P. Bellido, E. Crespo, L. Moliner, J. P. Rigla, M. J. Rodríguez-Álvarez, F. Sánchez, M. Seimetz, A. Soriano, L. F. Vidal, and J. M. Benlloch, "Minimization of border effects in monolithic scintillators using neural networks, based on MR-compatible SiPM arrays," vol. 1, no. 1, p. A19. [Online]. Available: https://doi.org/10.1186/2197-7364-1-S1-A19

[22] T. L. Laurence and S. X. Wang, "Dead pixel identification in positron emission tomography (PET)," U.S. Patent 9,841,515 B2 (WO 2015/145310 A1), Koninklijke Philips N.V., 2017, prioridad 28 mar. 2014.

[23] K. Gong, E. Berg, S. R. Cherry, and J. Qi, "Machine learning in PET: From photon detection to quantitative image reconstruction," *Proceedings of the IEEE*, vol. 108, no. 1, pp. 51–68, 2020.

[24] M. Decuyper, M. Stockhoff, S. Vandenberghe, and R. Van Holen, "Artificial neural networks for positioning of gamma interactions in monolithic PET detectors," *Physics in Medicine and Biology*, vol. 66, no. 7, p. 075001, 2021.

[25] G. Jaliparthi, P. F. Martone, A. V. Stolin, and R. R. Raylman, "Deep residual-convolutional neural networks for event positioning in a monolithic annular PET scanner," vol. 66, no. 14, p. 145008.

[26] L. A. Hidalgo-Torres, D. Pérez-Benito, R. Chil, P. Macias, and J. J. Vaquero, "Predicting 3D photon interaction in a hexagonal positron emission tomography detector: A deep learning approach," in *CASEIB 2020, XXXVIII Congreso Anual de la Sociedad Española de Ingeniería Biomédica: libro de actas*, R. Hornero Sánchez and Sociedad Española de Ingeniería Biomédica, Eds. Universidad de Valladolid, faltan las paginas: sacalas del PDF del libro de actas.

[27] Y. Li, X. Cheng, A. LaBella, W. Zhao, and A. H. Goldan, "Interleaved signal multiplexing readout in depth-encoding Prism-PET detectors," *Medical Physics*, vol. 50, no. 9, pp. 5327–5338, 2023, demultiplexado con DL evaluado por recuperacion del flood map.

[28] M. Niu, X. Jiang, S. Huang, Z. Chen, Y. Zhao, Y. Yang, W. Huang, H. Yang, X. Zhang, Z. Deng, and Z. Gu, *A CNN-based demultiplexing method for an interleaved signal multiplexing readout with a TOF-DOI brain PET detector*, pages: 1.

[29] F. Loignon-Houle, F. Pagano, and A. J. Gonzalez, "Multi-input interaction positioning and signal demultiplexing with deep learning in semi-monolithic detectors," vol. 10, no. 4, pp. 547–557. [Online]. Available: https://ieeexplore.ieee.org/document/11177553/

[30] S. Sarkar, X. Ye, G. Datta, and P. A. Beerel, "Fixpix: Fixing bad pixels using deep learning," 2024. [Online]. Available: https://arxiv.org/abs/2310.11637

[31] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 16 000–16 009.

[32] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, pp. 1024–1034.

[33] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," in *Proc. 34th Int. Conf. Machine Learning (ICML)*, 2017.

[34] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," *arXiv preprint*, 2016.

[35] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. 32nd Int. Conf. Machine Learning (ICML)*, 2015, pp. 448–456.

[36] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Int. Conf. Learning Representations (ICLR)*, 2018.

[37] G. Corso, L. Cavalleri, D. Beaini, P. Liò, and P. Veličković, "Principal neighbourhood aggregation for graph nets," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020.

[38] H. Gao and S. Ji, "Graph U-Nets," in *Proc. 36th Int. Conf. Machine Learning (ICML)*, 2019.

[39] C. Ying, T. Cai, S. Luo, S. Zheng, G. Ke, D. He, Y. Shen, and T.-Y. Liu, "Do transformers really perform bad for graph representation?" in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.

[40] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[41] P. J. Huber, "Robust estimation of a location parameter," *The Annals of Mathematical Statistics*, vol. 35, no. 1, pp. 73–101, 1964.

[42] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Int. Conf. Learning Representations (ICLR)*, 2019.

[43] ——, "SGDR: Stochastic gradient descent with warm restarts," in *Int. Conf. Learning Representations (ICLR)*, 2017.